SensoryGuard

IOT-BASED REAL-TIME DATA CENTER MONITORING SYSTEM USING ESP-32 NODE MCU WITH INTEGRATED SENSORS AND MOBILE APPLICATION

OPERATIONAL MANUAL

MACE TEAM

PROJECT TEAM

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SENSORYGUARD OPERATIONAL MANUAL

OVERVIEW:

The SensoryGuard Monitoring System is designed to help monitor server room conditions by detecting temperature, humidity, smoke, vibration, and flame in real-time. It uses an ESP32 microcontroller with sensors and sends data to an Android mobile and web app. This manual explains how to use the system, covering features like live monitoring, alerts, data logs, and customizable notifications.

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MAINTENANCE AND TROUBLESHOOTING

Explains how to install and navigate the mobile app, with details on features and user interface.

Routine Checks

- Clean sensors regularly and ensure proper placement.
- Check for loose connections or sensor wear.
- Monitor power supply stability.

Common Issues & Fixes

- **No sensor data:** Check wiring and ensure all components are powered.
- **Exclamation point of hardware display:** Close and turn off prototype for a few hours or open the next day (Will still work even if a character is replaced with "!").
  - You can choose to rest the prototype or continue to use it until it goes back to normal
- **Wi-Fi issues:** Re-enter network credentials using the WiFi Manager.
- **No notifications:** Make sure notification permissions are enabled on the mobile device.

SAFETY AND USAGE NOTES

- For indoor use only, intended specifically for monitoring server room environments.
- Always disconnect the adapter when modifying hardware connections.
- Note: The system is for monitoring purposes only and does not include automated responses like fans.

SYSTEM COMPONENTS

The SensoryGuard system includes these main following components:

1. Hardware:

- ESP32 Development Board (WiFi enabled)
- DHT22 Sensor (Temperature & Humidity)
- Flame Sensor Module (3-wired)
- Vibration Sensor Module (SW-420)
- Smoke Sensor (MQ-2)
- Buzzer (for smoke & fire alerts)
- Adapter (5V 3A)

2. Software:

- Arduino IDE (for microcontroller programming)
- Capacitor - APK Android Studio (for mobile app deployment)
- Vercel (Web App Deployment)
- Supabase (for real-time database and logs)

HARDWARE SETUP

Follow these steps to setup the SensoryGuard prototype.

Step 1: Plug in the Adapter

- Make sure to use the provided 5V 3A adapter. If replaced, ensure the new one is also 5V 3A.

Step 2: Connect to WiFi

- Let the system load until it displays "Connect to: SensoryGuard-P1" or "SensoryGuard-P2."
- Open your device's Wi-Fi settings and select the displayed network.

- Once connected, a “WiFi Manager” screen will appear.
- Tap **Configure**, then select and input the SSID and password of your desired network.
- After a few seconds, it should say “WiFi connected.” If not, try reconnecting.

Step 3: Successfully Connected

- You may now use the SensoryGuard mobile or web application.
- Real-time sensor data will be shown both on the LCD display and in the app.

APPLICATION GUIDE

Explains how to install and navigate the mobile app, with details on features and user interface.

1. Installation

- Install the SensoryGuard app (Android only) via the provided APK file.
- For iOS, desktop, or other devices, access the web app through your browser:

  https://sensory-guard.vercel.app/.

2. Features

- **Live Monitoring:** Displays real-time sensor data on the dashboard.
- **Notifications:** Push alerts are sent when hazards are detected.
- **Data Logging:** View historical sensor readings stored in the database.

- **Alert Customization:** Enable or disable the buzzer.
- **Controls:** Adjust temperature, humidity, and smoke threshold values.
- **Settings:** View connected network, access WiFi manager, and download a PDF copy of this manual.

3. User Interface

- A user-friendly layout featuring live values, color-coded alerts, intuitive navigation, and access to historical data.

SYSTEM OPERATION

Outlines how the system works after setup, including sensor monitoring and hazard response behavior.

- Once powered and connected to Wi-Fi, the system begins real-time monitoring.
- Sensor readings are displayed on the LCD and updated to the app interface.
- If any sensor crosses the defined threshold, the system activates the buzzer (smoke or flame), updates the LED status (for all sensors), and sends a notification to the app.
- All sensor data is stored and accessible via the logging feature on the app or web version.

ABOUT THE DEVELOPERS

- The **SensoryGuard** system was developed and prototyped by a team of student technopreneurs from **Polytechnic University of the Philippines – Manila**, under the **Bachelor of Science in Computer Engineering** program. This project was self-funded, showcasing our team's dedication to merging hardware and software into a functional safety solution. This operational manual and prototype demonstrate our collective technical capabilities and our commitment to improving environmental safety in sensitive spaces like server rooms.

Key Contributors:

Carandang, Kyle Kryzel R.

- Project Manager, Hardware Developer (building & code), UI/UX Designer, and Feature Planner
- She led the hardware assembly, designed the system interface, coordinated the overall workflow, and ensured each module aligned with the project's goals.

Elumba, William James R.

- Lead Software Developer and System Functionality Integrator
- He handled the core software logic, database integration, and system feature implementation across platforms.

For possible future upgrades, you may contact ***kylekryzelcarandang4@gmail.com*** to reach the developers.